# HW6

Tuesday, 8 April 2025 9:47 AM

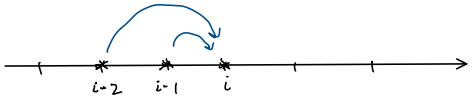
## Problem 1

(i) Derive difference operator for $\left(\frac{dT}{dx}\right)_{I_L}$ w/ $O(\Delta x^2)$ accuracy

(1) No. of terms to be kept in Taylor series

$N$ = order of derivative + order of accuracy
= 1 + 2
= 3

(2) Stencil




(3) N-1 truncated Taylor series about $T_{I_L}$

$$T_{I_L-1} = T_{I_L} - \left(\frac{dT}{dx}\right)_{I_L}\Delta x + \left(\frac{d^2T}{dx^2}\right)_{I_L}\frac{\Delta x^2}{2!} - O(\Delta x^3)$$

$$T_{I_L-2} = T_{I_L} - \left(\frac{dT}{dx}\right)_{I_L}(2\Delta x) + \left(\frac{d^2T}{dx^2}\right)_{I_L}\frac{(2\Delta x)^2}{2!} - O(\Delta x^3)$$

(4) Solving coupled linear equations

Subtracting the two terms:

$$T_{I_L-1} - T_{I_L-2} = \left(\frac{dT}{dx}\right)_{I_L}\Delta x - \left(\frac{3}{2}\Delta x^2\right)\left(\frac{d^2T}{dx^2}\right)_{I_L}$$

$$\left(\frac{d^2T}{dx^2}\right)_{I_L} = \left[(-T_{I_L-1} + T_{I_L-2}) + \left(\frac{dT}{dx}\right)_{I_L}\Delta x\right]\frac{2}{3(\Delta x^2)}$$

Substituting here

$$T_{I_L-1} = T_{I_L} - \left(\frac{dT}{dx}\right)_{I_L}\Delta x + \left(\frac{d^2T}{dx^2}\right)_{I_L}\frac{\Delta x^2}{2!} - O(\Delta x^3)$$

$$\therefore\ T_{I_L-1} = T_{I_L} - \left(\frac{dT}{dx}\right)_{I_L}\Delta x + \left[(-T_{I_L-1} + T_{I_L-2}) + \left(\frac{dT}{dx}\right)_{I_L}\Delta x\right]\frac{2}{3\Delta x^2}\cdot\frac{\Delta x^2}{2}$$

$$T_{I_L-1} = T_{I_L} - \left(\frac{dT}{dx}\right)_{I_L}\Delta x + \frac{1}{3}(-T_{I_L-1} + T_{I_L-2}) + \left(\frac{dT}{dx}\right)_{I_L}\left(\frac{1}{3}\Delta x\right)$$

$$T_{I_L-1} = T_{I_L} - \left(\frac{dT}{dx}\right)_{I_L}\Delta x + \frac{1}{3}\left(-T_{I_L-1} + T_{I_L-2}\right) + \left(\frac{dT}{dx}\right)_{I_L}\left(\frac{1}{3}\Delta x\right)$$

$$\left(\frac{dT}{dx}\right)_{I_L}\left(\Delta x - \frac{1}{3}\Delta x\right) = T_{I_L} - T_{I_L-1} - \frac{1}{3}T_{I_L-1} + \frac{1}{3}T_{I_L-2}$$

$$\left(\frac{dT}{dx}\right)_{I_L}\left(\frac{2}{3}\Delta x\right) = T_{I_L} - \frac{4}{3}T_{I_L-1} + \frac{1}{3}T_{I_L-2}$$

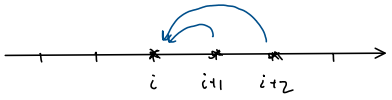
$$\left(\frac{dT}{dx}\right)_{I_L} = \left(\frac{3}{2\Delta x}\right)\left[\frac{3T_{I_L} - 4T_{I_L-1} + T_{I_L-2}}{3}\right]$$

$$\boxed{\left(\frac{dT}{dx}\right)_{I_L} = \frac{3T_{I_L} - 4T_{I_L-1} + T_{I_L-2}}{2\Delta x} + O(\Delta x^2)}$$

(ii) Following a similar strategy for calculating difference operator for $\left(\frac{dT}{dx}\right)_1$

(1) $N = 3$

(2)




(3) $$T_2 = T_1 + \left(\frac{dT}{dx}\right)_1\Delta x + \left(\frac{d^2T}{dx^2}\right)_1\frac{(\Delta x)^2}{2} + O(\Delta x^3)$$

$$T_3 = T_1 + \left(\frac{dT}{dx}\right)_1 2\Delta x + \left(\frac{d^2T}{dx^2}\right)_1\frac{(2\Delta x)^2}{2} + O(\Delta x^3)$$

(4) Subtracting the equations :

$$T_2 - T_3 = -\left(\frac{dT}{dx}\right)_1\Delta x - \left(\frac{d^2T}{dx^2}\right)_1\left(\frac{3}{2}\Delta x^2\right)$$

$$\left(\frac{d^2T}{dx^2}\right)_1 = \left[(T_3 - T_2) - \left(\frac{dT}{dx}\right)_1\Delta x\right]\frac{2}{3\Delta x^2}$$

Substituting

$$T_2 = T_1 + \left(\frac{dT}{dx}\right)_1\Delta x + \left(\frac{d^2T}{dx^2}\right)_1\frac{(\Delta x)^2}{2} + O(\Delta x^3)$$

$$T_2 = T_1 + \left(\frac{dT}{dx}\right)_1\Delta x + \left[(T_3 - T_2) - \left(\frac{dT}{dx}\right)_1\Delta x\right]\frac{\Delta x^2}{2}\cdot\frac{2}{3\Delta x^2}$$

$$\left(\frac{dT}{dx}\right)_1\left(-\frac{2\Delta x}{3}\right) = T_1 - T_2 + \frac{T_3}{3} - \frac{T_2}{3}$$

$$= \left[\frac{3T_1 - 4T_2 + T_3}{3}\right]\frac{3}{(-2\Delta x)}$$

$$\left(\frac{dT}{dx}\right)_1 = \frac{-3T_1 + 4T_2 - T_3}{2\Delta x} + O(\Delta x^2)$$

## Problem 2

4th order compact operator

$$c\frac{\partial U}{\partial x} = \nu\frac{\partial^2 U}{\partial x^2} \quad ; \quad @\ x=0,\ U=1 \quad @\ x=L=10,\ \frac{dU}{dx}=2$$

$$\downarrow$$

$$cF_i - \nu S_i = 0 \qquad \frac{cL}{\nu} = 2$$

$\left(\frac{\partial U}{\partial x}\right)_i$ $\left(\frac{\partial^2 U}{\partial x^2}\right)_i$

Compact differencing yields:

$$F_{i+1} + 4F_i + F_{i-1} = \frac{3}{\Delta x}(U_{i+1} - U_{i-1}) \quad \cdots (1)$$

Derived via operator approach

$$S_{i+1} + 10S_i + S_{i-1} = \frac{12}{\Delta x^2}(U_{i+1} - 2U_i + U_{i-1}) \quad \cdots (2)$$

We know: $cF_i = \nu S_i$

$S_i = \frac{c}{\nu}F_i$ → substituting in above equation

$$\therefore \quad \frac{c}{\nu}F_{i+1} + \frac{10c}{\nu}F_i + \frac{c}{\nu}F_{i-1} = \frac{12}{\Delta x^2}(U_{i+1} - 2U_i + U_{i-1}) \quad \cdots (3)$$

Similarly, $F_i = \frac{\nu}{c}S_i$

$$\frac{\nu}{c}S_{i+1} + 4S_i + S_{i-1} = \frac{3}{\Delta x}(U_{i+1} - U_{i-1}) \quad \cdots (4)$$

We can combine (1) (3) in matrix form

From (1)

$$F_{i+1} + 4F_i + F_{i-1} - \frac{3}{\Delta x}(U_{i+1} - U_{i-1}) = 0$$

$$\Rightarrow \left[F_{i+1} - \frac{3}{\Delta x}U_{i+1}\right] + 4F_i + \left[\frac{3}{\Delta x}U_{i-1} + F_{i-1}\right] = 0$$

From (3)

$$\frac{c}{\nu}F_{i+1} + \frac{10c}{\nu}F_i + \frac{c}{\nu}F_{i-1} = \frac{12}{\Delta x^2}(U_{i+1} - 2U_i + U_{i-1})$$

$$= \left[\frac{c}{\nu} F_{i+1} - \frac{12}{\Delta x^2} U_{i+1}\right] + \left[10\frac{c}{\nu} F_i + \frac{24}{\Delta x^2} U_i\right] + \left[\frac{c}{\nu} F_{i-1} - \frac{12}{\Delta x^2} U_{i-1}\right] = 0$$

Combining above expressions in matrix form:

$$\underbrace{\begin{bmatrix} -\frac{3}{\Delta x} & 1 \\ -\frac{12}{\Delta x^2} & c/\nu \end{bmatrix}}_{L_i} \begin{bmatrix} U_{i+1} \\ F_{i+1} \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 4 \\ \frac{24}{\Delta x^2} & 10\frac{c}{\nu} \end{bmatrix}}_{M_i} \begin{bmatrix} U_i \\ F_i \end{bmatrix} + \underbrace{\begin{bmatrix} 3/\Delta x & 1 \\ -12/\Delta x^2 & c/\nu \end{bmatrix}}_{N_i} \begin{bmatrix} U_{i-1} \\ F_{i-1} \end{bmatrix} = \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}_{d_i}$$

If we define $V_i = \begin{bmatrix} U_i \\ F_i \end{bmatrix}$ ; where $V_{i-1} = \begin{bmatrix} U_{i-1} \\ F_{i-1} \end{bmatrix}$ we obtain:

$$\boxed{L_i V_{i-1} + M_i V_i + N_i V_{i+1} = d_i} \quad \cdots (5)$$

↳ Solving this system of equations will yield $O(\Delta x^4)$ accurat solution

Apply Boundary conditions

Interior points

$L_i V_{i-1} + M_i V_i + N_i V_{i+1} = d_i$

↳ One-sided differencing that relates $\frac{dU}{dx}$ at the boundary to two neighboring points

@ $x = 0$ $(i=1)$ $U = 1$ $\quad V_1 = \begin{bmatrix} 1 \\ F_1 \end{bmatrix}$

$L_2 V_1 + M_2 V_2 + N_2 V_3 = d_1$

@ $x = L$ $\frac{dU}{dx} = F = 2$ $\quad V_{IL} = \begin{bmatrix} U_{IL} \\ 2 \end{bmatrix}$

$L_{IL-1} V_{IL-2} + M_{IL-1} V_{IL-1} + N_{IL} V_{IL} = d$

## Overall algorithm

1. Discretize the domain ( $x_i = (i-1)\Delta x$ ; $i = 1, 2, 3 \cdots n, n+1$ )
   ↳ grid is initialized for given $\Delta x$
2. Use compact differencing to discretize ODE (arriving at eq (5) from above)
3. Apply given boundary conditions & solve for interior points
   → Iterate using for loop
4. Plot and analyze results